

Team Responsibilities Division

Detailed module ownership, functional tasks, and integration assignments

OVERALL MODULE OWNERSHIP

MEMBER	MAIN MODULE
Ama Meena + Navin	Interactive CLI & Register File
Prateet	Pipeline Simulator
Navin + Aman	CPU Execution & Debugger + ISA/Assembler/Disassembler
Prateet + Chirag	Memory + Control/Hazard Units
Amit + Prateet	ISA + Integration & Testing

1. Interactive CLI & Register File

Ama Meena + Navin

- Develop the interactive command-line interface.
- Implement the register file.
- Provide commands to view and modify registers.
- Implement the instruction execution interface.
- Display CPU/register state through the CLI.
- Connect CLI commands with the CPU simulator.

2. Pipeline Simulator

Prateet

- Implement the CPU pipeline.
- Implement IF, ID, EX, MEM, and WB stages.
- Implement pipeline registers.
- Handle instruction movement between pipeline stages.
- Implement pipeline stalls and flushes.
- Implement pipeline visualization/status display.
- Integrate pipeline execution with the CPU.

3. Step-by-Step CPU Execution & Debugger

Navin + Aman

- Implement step-by-step CPU execution.
- Implement step, run, continue, and related execution commands.
- Implement breakpoints and debugging controls.
- Display PC, registers, pipeline state, and execution status.
- Implement cycle-by-cycle execution.
- Provide an interactive CLI debugger.

4. Memory & Control/Hazard Units

Prateet + Chirag

- Implement instruction and data memory.
- Implement load/store operations.
- Develop the control unit.
- Generate control signals for instruction execution.
- Implement hazard detection and handling.
- Implement branch hazard/flush logic.
- Implement stall and forwarding-related control where required.
- Integrate interrupts and exceptions with the control system.

5. ISA, Assembler/Disassembler & Integration

Amit + Prateet

- Implement isa.cpp and isa.h.
- Define instruction formats and opcodes.
- Implement instruction encoding and decoding.
- Implement assembler/disassembler functionality.
- Integrate ISA decoding with the pipeline and CPU.
- Perform system-level integration testing.
- Test interaction between the ISA, pipeline, memory, and debugger.
- Identify and fix integration issues.